Dr. Eliette L. Reboud and Dr. Fabien L. Condamine

March 03rd, 2026

Dear Editors-in-Chief and Associate Editors,

Please find enclosed our manuscript entitled "**Laurasian origin and asymmetric regional diversification in the tropics of Troidini butterflies revealed by whole-genome phylogenomics**" which we submit for evaluation and publication in *Systematic Biology*.

One of the central challenges of evolutionary biology is understanding how global biodiversity arises and why it is unevenly distributed across regions and clades. To address this question, we must integrate large-scale genomic resources with models that explicitly link diversification processes to ecological and historical drivers. Although insects are the most diverse organisms on Earth, genomic-scale phylogenetic frameworks remain scarce for pantropical lineages. Here, we aim to further discuss the origins of tropical richness and test whether the same evolutionary processes govern biodiversity in different tropical regions.

We focus on the butterfly tribe Troidini (Papilionidae), a pantropical lineage renowned for its charismatic species, such as birdwing butterflies, and their long-standing ecological relationship with the toxic Aristolochiaceae plant family. They exhibit striking disparities in species richness across continents, thriving in the Neotropics, Indo-Australian tropics, and Madagascar but absent in Africa. This clade provides a unique system to explore how macroevolutionary processes shape tropical diversity in different parts of the world. Despite their prominence, hypotheses about their origin and diversification have been speculative due to the absence of a species-level phylogenetic framework.

To address this challenge, we generated the most extensive genomic datasets ever compiled for an insect radiation. Using a combination of long- and short-read sequencing, we produced five reference genomes and assembled the complete genomes of 133 species, representing approximately 96% of the known Troidini species. Using thousands of nuclear loci and mitochondrial data, we inferred robust, time-calibrated phylogenies. Then, we applied cutting-edge macroevolutionary approaches, including Bayesian relaxed-clock analyses, process-based models of range evolution, and diversification models that explicitly incorporate environmental and diversity-dependent drivers. This level of taxonomic and genomic coverage is rare in entomology and allows us to rigorously test macroevolutionary hypotheses that are typically inaccessible in pantropical insect clades.

Our analyses reveal that the Troidini originated in Laurasia during the mid-Eocene. They dispersed southward concurrently into the Neotropics and the Indo-Australian archipelago, experiencing local extinctions at high latitudes along the way. Notably, diversification proceeded asymmetrically across the tropics. Southeast Asian lineages exhibited high speciation rates tied to sea-level fluctuations and biogeographic dynamics in an island setting. In contrast, Neotropical lineages followed diversity-dependent trajectories influenced by range dynamics, with dispersal and local extinction playing important roles. These findings demonstrate that regional tropical biodiversity can be driven by fundamentally distinct evolutionary processes. Our work challenges the long-held assumption of uniformity in tropical diversification and offers new insights into how large-scale climatic and geographic processes interact to shape global insect richness.

We believe this manuscript will resonate strongly with readers of *Systematic Biology*. It presents one of the most robust genomic datasets for a pantropical insect clade and integrates it with advanced evolutionary models to reveal novel, generalizable insights into the uneven assembly of tropical biodiversity. Our study contributes to the broader comparative framework for understanding why the tropics are so diverse and how that diversity is distributed unevenly across regions.

Yours sincerely,

Dr. Eliette L. Reboud and Dr. Fabien L. Condamine, on behalf of our co-authors

Dr. Fabien L. Condamine
CNRS, Institut des Sciences de l'Evolution de Montpellier, Place Eugène Bataillon, 34095 Montpellier, France.
Phone: +33 674 932 296 | Email: fabien.condamine@gmail.com